

of reaching 100 GW nuclear capacity by 2047, and nuclear power is also set to play a key role in the nation's decarbonisation strategy, as companies look to replace their fossil fuel-based captive power plants.

In mid-February this year, the country's top nuclear scientists met over 30 industry leaders. The meeting was attended by executives from Adani Group, Reliance, Tata, JSW, Jindal Steel, Vedanta, Hindalco Industries, as well as some renewable energy companies. As part of the discussions, which focused on investment opportunities in the nuclear sector, several operational issues were flagged.

ON THE TABLE

The nearly three-hour meeting at the NPCIL headquarters in Mumbai saw a free-wheeling discussion between industry players and nuclear scientists led by NPCIL Chairman and Managing Director Bhuwan Chandra Pathak. This was the first meeting after Finance Minister Nirmala Sitharaman launched the Nuclear Energy Mission and announced that the government will amend laws to open doors for private sector participation in the sector.

NPCIL briefed the companies on the steps it's taking to make it easier for private players to invest in the sector. Private players, on their part, pointed out the roadblocks. One of the biggest issues that they raised was the civil liability law related to financial responsibility in case of a nuclear accident. Another concern was the high operation and maintenance costs that could eat into their profitability.

The companies also sought more control over the nuclear plants they will invest in; at present, the control rests with NPCIL. The corporation operates 25 nuclear power reactors at seven sites across the nation with an installed capacity of 8.8 GW. It is set to add another 15 GW by 2032. Since the sector has strict safety regulations to protect public health,

NUCLEAR ROADMAP



**100 GW OF
NUCLEAR
POWER BY
2047 FROM THE
PRESENT 8.1 GW**



**₹20,000 CRORE
NUCLEAR
ENERGY
MISSION
ANNOUNCED
BY THE
GOVERNMENT**



**PARTNERSHIP
WITH PRIVATE
PLAYERS FOR
CAPTIVE
USE OF
NUCLEAR
POWER**



**AMENDMENTS
TO THE ATOMIC
ENERGY ACT
AND CIVIL
LIABILITY FOR
NUCLEAR
DAMAGE ACT
ARE BEING
UNDERTAKEN**

environment and the integrity of the plants, while keeping India in line with international standards, industry felt the need for viability gap funding (VGF) and power purchase agreements (PPAs). Industry wants PPAs for nuclear power to be on the lines of agreements for renewable power so that they can cover the high upfront costs of nuclear plants.

There's the also the issue of uranium availability. India imports uranium for nuclear plants due to the poor quality of the ore found domestically. The government has assured adequate availability of the metal. "As we understand, there are adequate fuel provisions, both through indigenous as well as imported means, to support the existing and planned nuclear power programme," Pathak tells *Business Today*. Anil Kakodar, Chancellor, Homi Bhabha National Institute, tells *BT* that India should focus on using a combination of uranium and thorium due to the abundance of the latter in the country. The used fuel from the reactors could be recycled to set up additional power generation power capacity.

The government needs to address these concerns for private investments to flow in and India's nuclear energy efforts to take off.

AMBITIOUS GOALS

The government's decision to allow private capital in the nuclear sector has generated huge interest. The NPCIL floated a request for proposal (RFP) on December 31, 2024, inviting Indian companies to participate in setting up 220 MW BSRs with upgraded pressurised heavy water reactors (PHWR) technology for captive power generation. It is a proven technology that can be scaled up at a faster pace compared to the newer technologies under development.

The fact that PHWRs have been operating in India for decades is a big advantage. Unlike newer reactor designs, which are still under development or need more regulatory approvals, the 220 MW BSRs work on a tried-and-test-